CS4500 Operating Systems I
Project 3
Memory Management

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Statement: "I have neither given nor received unauthorized assistance on this work and am in compliance with Student Academic Ethics Code Policy."

AI usage disclosure: I have used Gen AI tool as a learning assistant in this work
Name of the AI tool: chatGPT4.0
Link of the Conversation: https://chatgpt.com/c/69ea5b1a-9394-83e8-82d4-182cf57220a8

Task 1

1.1 How I Solved the Problem

The problem is determining how much virtual address space a process is using inside the Linux kernel. User-level tools like pmap can show memory usage, but this assignment specified to access that information directly from the kernel.

To solve this, I wrote a kernel module that takes a process ID as input and locates the corresponding task_struct. From there, I accessed the process's memory descriptor (mm_struct), which contains all virtual memory regions. I then traversed these regions using the kernel's maple tree (mm->mm_mt) and summed the size of each virtual memory area (VMA). This allowed me to compute both the total virtual address space and the number of VMAs for the process. The results are printed to the kernel log using printk.

1.2 3 facts about my code

The module calculates the total virtual address space of a process by iterating through all of its virtual memory areas (VMAs) using the Linux kernel's maple tree structure (mm->mm_mt)

It accepts a process ID as a parameter (pid) at load time, allowing it to dynamically analyze different processes without modifying the code.

The module safely handles edge cases, like when a PID does not exist or when the target process is a kernel thread, by returning appropriate error codes and logging messages with printk.

1.3 How to Run My Code

To run the code:

```
cd task1/
make
Sudo insmod VA_space.ko pid=<PID> # can get PID with pgrep bash
Sudo dmesg | tail -20
Sudo rmmod VA_space
```

1.4 Output

```
ubuntu@ip-172-31-69-149:~/OS/task2$ cd ./../task1/
ubuntu@ip-172-31-69-149:~/OS/task1$ make all
make -C /lib/modules/6.17.0-1010-aws/build M=/home/ubuntu/OS/task1 modules
make[1]: Entering directory '/usr/src/linux-headers-6.17.0-1010-aws'
make[2]: Entering directory '/home/ubuntu/OS/task1'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  You are using:           gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  CC [M]  VA_space.o
  MODPOST Module.symvers
  CC [M]  VA_space.mod.o
  CC [M]  .module-common.o
  LD [M]  VA_space.ko
  BTF [M] VA_space.ko
Skipping BTF generation for VA_space.ko due to unavailability of vmlinux
make[2]: Leaving directory '/home/ubuntu/OS/task1'
make[1]: Leaving directory '/usr/src/linux-headers-6.17.0-1010-aws'
ubuntu@ip-172-31-69-149:~/OS/task1$ pgrep bash
1758
ubuntu@ip-172-31-69-149:~/OS/task1$ sudo insmod VA_space.ko pid=1758
ubuntu@ip-172-31-69-149:~/OS/task1$ dmesg | tail -20
dmesg: read kernel buffer failed: Operation not permitted
ubuntu@ip-172-31-69-149:~/OS/task1$ sudo dmesg | tail -20
[  649.469269] va_space: VMA #29: 0x70ef7189a000 - 0x70ef7189b000  size=4096 bytes
[  649.469270] va_space: VMA #30: 0x70ef7189b000 - 0x70ef7189c000  size=4096 bytes
[  649.469272] va_space: VMA #31: 0x70ef7189c000 - 0x70ef7189d000  size=4096 bytes
[  649.469273] va_space: VMA #32: 0x70ef7189d000 - 0x70ef7189e000  size=4096 bytes
[  649.469275] va_space: VMA #33: 0x70ef7189e000 - 0x70ef7189f000  size=4096 bytes
[  649.469276] va_space: VMA #34: 0x70ef7189f000 - 0x70ef718a1000  size=8192 bytes
[  649.469278] va_space: VMA #35: 0x70ef718a1000 - 0x70ef718a5000  size=16384 bytes
[  649.469279] va_space: VMA #36: 0x70ef718a5000 - 0x70ef718a7000  size=8192 bytes
[  649.469281] va_space: VMA #37: 0x70ef718a7000 - 0x70ef718a9000  size=8192 bytes
[  649.469283] va_space: VMA #38: 0x70ef718a9000 - 0x70ef718aa000  size=4096 bytes
[  649.469284] va_space: VMA #39: 0x70ef718aa000 - 0x70ef718d5000  size=176128 bytes
[  649.469286] va_space: VMA #40: 0x70ef718d5000 - 0x70ef718df000  size=40960 bytes
[  649.469287] va_space: VMA #41: 0x70ef718df000 - 0x70ef718e1000  size=8192 bytes
[  649.469289] va_space: VMA #42: 0x70ef718e1000 - 0x70ef718e3000  size=8192 bytes
[  649.469290] va_space: VMA #43: 0x7ffcd5a2f000 - 0x7ffcd5a50000  size=135168 bytes
[  649.469292] va_space: Process 'bash' (PID 1758)
[  649.469293] va_space:   Number of VMAs : 43
[  649.469294] va_space:   Total VA space  : 9420800 bytes
[  649.469295] va_space:   Total VA space  : 9200 kB
[  649.469296] va_space: (Compare with: sudo pmap 1758 | tail -1)
ubuntu@ip-172-31-69-149:~/OS/task1$ sudo pmap 1758 | tail -1
 total             9204K
ubuntu@ip-172-31-69-149:~/OS/task1$ sudo rmmod VA_space
```

Task 2

2.1 How I Solved the Problems

The goal of this task was to determine whether a given virtual address in a specific process is currently mapped in physical memory or has been swapped out to disk. To solve this, I implemented a Linux kernel module that takes both a process ID and a virtual address as input parameters.

First, I converted the input virtual address from a hex string into an unsigned long using kstrtoul, since sscanf is not available in kernel space. I then located the target process using pid_task(find_get_pid(pid)) and accessed its memory descriptor (mm_struct). From there, I performed a full page table walk through the five levels of the Linux page table hierarchy (PGD → P4D → PUD → PMD → PTE).

At each level, I checked whether the entry was valid before continuing. Once the PTE was reached, I used pte_present() to determine whether the page is currently in memory or has been swapped out. Finally, I logged the result to the kernel log using printk.

2.2 3 facts about my code

The code performs a manual 5-level page table walk, following the Linux virtual memory translation process from PGD down to PTE.

It determines whether a virtual address is in RAM or swapped to disk by checking the present bit in the PTE, and extracts the physical frame number PFN if the page is in memory.

The module safely converts user input and handles invalid cases such as missing processes, kernel threads, and unmapped addresses at any stage of the page table traversal, for safe accesss

2.3 How to Run Our Code

To run the code

```
cd task2/
make
PID=$(pgrep bash | head -1)
sudo pmap $PID # to get PID and see hex address
sudo insmod VA_status.ko pid=<PID> vaddr=<HEX_ADDR>
dmesg | tail -20
sudo rmmod VA_status
```

2.4 Output

```
ubuntu@ip-172-31-69-149:~/OS$ cd task2/
ubuntu@ip-172-31-69-149:~/OS/task2$ make all
make -C /lib/modules/6.17.0-1010-aws/build M=/home/ubuntu/OS/task2 modules
make[1]: Entering directory '/usr/src/linux-headers-6.17.0-1010-aws'
make[2]: Entering directory '/home/ubuntu/OS/task2'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  You are using:           gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  CC [M]  VA_status.o
  MODPOST Module.symvers
  CC [M]  VA_status.mod.o
  CC [M]  .module-common.o
  LD [M]  VA_status.ko
  BTF [M] VA_status.ko
Skipping BTF generation for VA_status.ko due to unavailability of vmlinux
make[2]: Leaving directory '/home/ubuntu/OS/task2'
make[1]: Leaving directory '/usr/src/linux-headers-6.17.0-1010-aws'
ubuntu@ip-172-31-69-149:~/OS/task2$ PID=$(pgrep bash | head -1)
ubuntu@ip-172-31-69-149:~/OS/task2$ sudo pmap $PID
1758:   -bash
000063995af54000    192K r---- bash
000063995af84000    956K r-x-- bash
000063995b073000    212K r---- bash
000063995b0a8000     16K r---- bash
000063995b0ac000     36K rw--- bash
000063995b0b5000     44K rw---   [ anon ]
000063996c13c000   1584K rw---   [ anon ]
000070ef71200000   2988K r---- locale-archive
000070ef715a7000    356K r---- LC_CTYPE
000070ef71600000    160K r---- libc.so.6
000070ef71628000   1568K r-x-- libc.so.6
000070ef717b0000    316K r---- libc.so.6
000070ef717ff000     16K r---- libc.so.6
000070ef71803000      8K rw--- libc.so.6
000070ef71805000     52K rw---   [ anon ]
000070ef71856000      4K r---- LC_NUMERIC
000070ef71857000      4K r---- LC_TIME
000070ef71858000      4K r---- LC_COLLATE
000070ef71859000      4K r---- LC_MONETARY
000070ef7185a000      4K r---- SYS_LC_MESSAGES
000070ef7185b000     28K r--s- gconv-modules.cache
000070ef71862000     12K rw---   [ anon ]
000070ef71865000     56K r---- libtinfo.so.6.4
000070ef71873000     76K r-x-- libtinfo.so.6.4
000070ef71886000     56K r---- libtinfo.so.6.4
000070ef71894000     16K r---- libtinfo.so.6.4
000070ef71898000      4K rw--- libtinfo.so.6.4
000070ef71899000      4K r---- LC_PAPER
000070ef7189a000      4K r---- LC_NAME
000070ef7189b000      4K r---- LC_ADDRESS
000070ef7189c000      4K r---- LC_TELEPHONE
000070ef7189d000      4K r---- LC_MEASUREMENT
```

```
000070ef71898000      4K rw--- libtinfo.so.6.4
000070ef71899000      4K r---- LC_PAPER
000070ef7189a000      4K r---- LC_NAME
000070ef7189b000      4K r---- LC_ADDRESS
000070ef7189c000      4K r---- LC_TELEPHONE
000070ef7189d000      4K r---- LC_MEASUREMENT
000070ef7189e000      4K r---- LC_IDENTIFICATION
000070ef7189f000      8K rw---   [ anon ]
000070ef718a1000     16K r----   [ anon ]
000070ef718a5000      8K r----   [ anon ]
000070ef718a7000      8K r-x--   [ anon ]
000070ef718a9000      4K r---- ld-linux-x86-64.so.2
000070ef718aa000    172K r-x-- ld-linux-x86-64.so.2
000070ef718d5000     40K r---- ld-linux-x86-64.so.2
000070ef718df000      8K r---- ld-linux-x86-64.so.2
000070ef718e1000      8K rw--- ld-linux-x86-64.so.2
00007ffcd5a2f000    132K rw---   [ stack ]
ffffffffff600000      4K --x--   [ anon ]
 total             9204K
ubuntu@ip-172-31-69-149:~/OS/task2$ sudo insmod VA_status.ko pid=1758 vaddr=000063995af54000
ubuntu@ip-172-31-69-149:~/OS/task2$ sudo dmesg | tail -20
[  649.469286] va_space: VMA #40: 0x70ef718d5000 - 0x70ef718df000  size=40960 bytes
[  649.469287] va_space: VMA #41: 0x70ef718df000 - 0x70ef718e1000  size=8192 bytes
[  649.469289] va_space: VMA #42: 0x70ef718e1000 - 0x70ef718e3000  size=8192 bytes
[  649.469290] va_space: VMA #43: 0x7ffcd5a2f000 - 0x7ffcd5a50000  size=135168 bytes
[  649.469292] va_space: Process 'bash' (PID 1758)
[  649.469293] va_space:   Number of VMAs : 43
[  649.469294] va_space:   Total VA space  : 9420800 bytes
[  649.469295] va_space:   Total VA space  : 9200 kB
[  649.469296] va_space: (Compare with: sudo pmap 1758 | tail -1)
[  702.294240] va_space: Module unloaded by process 'rmmod' (PID 3020)
[  895.286106] va_status: Module loaded by process 'insmod' (PID 3349)
[  895.286112] va_status: Target PID=1758, virtual address=0x63995af54000 (109510307168256)
[  895.286115] va_status: Found process 'bash' (PID 1758)
[  895.286117] va_status: PGD entry found at ffff8b8fc410e638, value=0x8000000009a8e067
[  895.286119] va_status: P4D entry found
[  895.286120] va_status: PUD entry found
[  895.286122] va_status: PMD entry found
[  895.286123] va_status: PTE entry found at ffff8b8fc6d8baa0, value=0x800000002dfdd005
[  895.286124] va_status: Address 0x63995af54000 is IN MEMORY (PRESENT bit set)
[  895.286126] va_status:   The physical page frame number (PFN): 188381
ubuntu@ip-172-31-69-149:~/OS/task2$ sudo rmmod VA_status
```